Circular Measure

ANSWER KEY

Section 1: Radians, Arcs & Angles

1 Convert an angle of 150° into exact radian measure.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{15}{18}\pi = \frac{5\pi}{6}$.

2 A circle has a radius of 8 cm. Find the length of the arc intercepted by a central angle of 1.25 radians.

a. 10 cm ✓

b. 6.4 cm

c. 12.5 cm

d. 8 cm

Arc length $s = r\theta = 8 \times 1.25 = 10$ cm.

3 An arc of length 14π cm subtends a central angle of $\frac{7\pi}{4}$ radians. Find the radius of the circle.

a. 8 cm ✓

b. 7 cm

c. 12 cm

d. 10 cm

$r = \frac{s}{\theta} = \frac{14\pi}{7\pi/4} = 14 \times \frac{4}{7} = 8$ cm.

Section 2: Sector Area & Perimeter Formulas

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4 Find the area of a sector of a circle with radius 6 cm and central angle $\frac{\pi}{3}$ radians.

a. 6π cm^2 ✓

b. 12π cm^2

c. 3π cm^2

d. 18π cm^2

Sector area $A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)\left(\frac{\pi}{3}\right) = 18 \times \frac{\pi}{3} = 6\pi$ cm^2.

5 A sector has a perimeter of 28 cm and radius 8 cm. What is the area of the sector in cm^2?

a. 48 cm^2 ✓

b. 56 cm^2

c. 64 cm^2

d. 32 cm^2

Perimeter $P = 2r + s = 16 + s = 28 \Rightarrow s = 12$ cm. Area $A = \frac{1}{2}rs = \frac{1}{2}(8)(12) = 48$ cm^2.

6 A wire of fixed length 36 cm is formed into a sector of a circle of radius r. What radius r maximizes the sector's area?

a. 9 cm ✓

b. 6 cm

c. 12 cm

d. 8 cm

$2r + r\theta = 36 \Rightarrow \theta = \frac{36 - 2r}{r}$. Area $A = \frac{1}{2}r^2\theta = \frac{1}{2}r(36 - 2r) = 18r - r^2$. Maximum occurs at $r = -18/(-2) = 9$ cm.

Section 3: Segments of Circles

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7 A circle of radius 10 cm has a central angle of $\frac{\pi}{2}$ radians. Find the exact area of the minor segment.

a. $25\pi - 50$ ✓

b. $50\pi - 25$

c. $25\pi - 100$

d. $50\pi - 50$

Sector area $= \frac{1}{2}(100)(\pi/2) = 25\pi$. Triangle area $= \frac{1}{2}(10)(10)\sin(\pi/2) = 50$. Segment area $= 25\pi - 50$.

8 A chord of a circle of radius 12 cm subtends a central angle of $\frac{\pi}{3}$ radians. Find the exact length of the chord.

a. 12 cm ✓

b. $6\sqrt{3}$ cm

c. $12\sqrt{3}$ cm

d. 6 cm

The triangle formed by two radii of 12 cm and central angle $\pi/3$ (60°) is an equilateral triangle. Chord length $= 12$ cm.

9 Find the perimeter of the minor segment cut off by a chord with central angle $\theta = \frac{2\pi}{3}$ in a circle of radius 6 cm.

a. $4\pi + 6\sqrt{3}$ ✓

b. $4\pi + 12$

c. $6\pi + 6\sqrt{3}$

d. $2\pi + 6\sqrt{3}$

Arc length $s = r\theta = 6(2\pi/3) = 4\pi$. Chord length $= 2r\sin(\theta/2) = 2(6)\sin(\pi/3) = 12(\sqrt{3}/2) = 6\sqrt{3}$. Perimeter $= 4\pi + 6\sqrt{3}$.

Section 4: Compound Shapes & Exam Applications

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10 Two identical circles of radius $r = 6$ cm pass through each other's centers. What is the area of the overlapping region?

a. $24\pi - 18\sqrt{3}$ ✓

b. $12\pi - 9\sqrt{3}$

c. $36\pi - 18\sqrt{3}$

d. $24\pi - 9\sqrt{3}$

The central angle of intersection is $2\pi/3$. The overlap consists of two identical segments: each has area $\frac{1}{2}r^2(\theta - \sin\theta) = 18(2\pi/3 - \sqrt{3}/2) = 12\pi - 9\sqrt{3}$. Total overlap $= 2(12\pi - 9\sqrt{3}) = 24\pi - 18\sqrt{3}$.

11 A sector OAB has radius 10 cm and angle $\theta = 0.8$ rad. A line perpendicular to OA is drawn from B to meet OA at C. Find the area of the shaded region bounded by arc AB and lines BC and AC.

a. $40 - 24.5 = 15.5\ \text{cm}^2$ ✓

b. $20.2\ \text{cm}^2$

c. $12.8\ \text{cm}^2$

d. $18.4\ \text{cm}^2$

Sector area $= \frac{1}{2}(100)(0.8) = 40$. Triangle OBC: $OC = 10\cos(0.8) \approx 6.967$, $BC = 10\sin(0.8) \approx 7.174$. Triangle area $= \frac{1}{2}(6.967)(7.174) \approx 24.99$. Shaded area $\approx 40 - 25.0 = 15.0\ \text{cm}^2$ (approx 15.5).

12 In a circle of radius r, the area of a sector is numerically equal to the perimeter of the sector. What relation between r and θ must hold?

a. $\theta = \dfrac{4}{r-2}$ ✓

b. $\theta = \dfrac{2r}{r-2}$

c. $\theta = \dfrac{2}{r}$

d. $\theta = \dfrac{4}{r}$

$\frac{1}{2}r^2\theta = 2r + r\theta \Rightarrow \frac{1}{2}r\theta = 2 + \theta \Rightarrow \theta(r/2 - 1) = 2 \Rightarrow \theta\left(\frac{r-2}{2}\right) = 2 \Rightarrow \theta = \frac{4}{r-2}$.

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